Problem Set 5

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```r
seed_string <- "Blair, Thomas, Aaron, John"
seed_number <-
  seed_string |>
  rlang::hash() |> # hash the string to a hex number
  substr(1, 8) |> # make the number smaller
  strtoi(base = 16) # convert to decimal

set.seed(seed_number)
```

Hamilton's Problems

Exercise 3.1

Exercise 3.2

We can rewrite Y_t in the form

$$\xi_t = \mathbf{F}\xi_{t-1} + \eta_t.$$

Then,

$$\begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix} = \begin{bmatrix} 1.1 & -0.18 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ 0 \end{bmatrix}$$

We find the eigenvalues of $\mathbf{F}$ by solving

$$\begin{aligned} \det \begin{pmatrix} 1.1-\lambda & -0.18 \\ 1 & -\lambda \end{pmatrix} &= 0 \\ -1.1\lambda + \lambda^2 + 0.18 &= 0 \\ \lambda^2 - 1.1\lambda + 0.18 &= 0 \\ (\lambda - 0.9)(\lambda - 0.2) &= 0. \end{aligned}$$

Therefore, the two eigenvalues are $\lambda_1 = 0.2$ and $\lambda_2 = 0.9$. Because

$$max_i|\lambda_i| < 1$$

we know that this AR(2) process admits a covariance-stationary MA(∞) process.

To calculate the mean and autocovariance, we proceed with the eigendecomposition of $\mathbf{F}$, which is:

$$\mathbf{T\Lambda T}^{-1} = \begin{pmatrix} 0.2 & 0.9 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0.2 & 0 \\ 0 & 0.9 \end{pmatrix} \begin{pmatrix} -\frac{10}{7} & \frac{9}{7} \\ \frac{10}{7} & -\frac{2}{7} \end{pmatrix}$$

Then,

$$\begin{aligned} F_{1,1}^j &= 0.2^2 \left(-\frac{10}{7}\right) + 0.9^2 \left(\frac{10}{7}\right) \\ &= \left(\frac{-2}{7}\right) 0.2^j + \left(\frac{9}{7}\right) 0.9^j. \end{aligned}$$

*Authors listed alphabetically by and contributed equally to every question.

If $\psi_j = \left(\frac{-2}{7}\right) 0.2^j + \left(\frac{9}{7}\right) 0.9^j$, then

$$Y_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}.$$

We can see that

$$\mathbb{E}[Y_t] = 0.$$

For the autocovariance,

$$\begin{aligned}
\gamma_j &= \mathbb{E}[(Y_t - \mu)(Y_{t-k} - \mu) \\
&= \mathbb{E}(Y_t Y_{t-k}) = \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}\right)\left(\sum_{l=0}^{\infty} \psi_l \varepsilon_{t-l}\right)\right] \\
&= \mathbb{E}\left[\sum_{n=0}^{\infty}\sum_{j=0}^{n} \left(\psi_j \varepsilon_{t-j}\right)\left(\psi_{n-j}\varepsilon_{t-(n-j)}\right)\right] \text{(by Cauchy product)} \\
&= \sum_{n=0}^{\infty}\sum_{j=0}^{n} \psi_j \psi_{n-j} \mathbb{E}[\varepsilon_{t-j}\varepsilon_{t-n+j}]
\end{aligned}$$

We know that $\mathbb{E}[\varepsilon_{t-j}\varepsilon_{t-n+j}] = 1$ when the indexes $t - j = t - n + j$, which is when $n = 2j$. Suppose $n = 0$, then $j = 0$ and we have $\psi_0\psi_0$. Suppose $n = 2$, then $j = 1$ and we have $\psi_1\psi_1$.

Thus, we can observe that the above expression simplifies to

$$\begin{aligned}
\gamma_j &= \sum_{j=0}^{\infty} \psi_j^2 \mathbf{1}\{j \text{ is even}\} \\
&= \sum_{j=0}^{\infty} \left[\left(-\frac{2}{7}\right) 0.2^j + \left(\frac{9}{7}\right) 0.9^j\right]^2 \mathbf{1}\{j \text{ is even}\}
\end{aligned}$$

Excercise 3.3

Excercise 11.5

The Permanent Income Model

1. Finding the Autocovariances

2. The Vector Process

3. Simulation Exercise

The Kalman Filter

1. Analytical Moments

We are given the ARMA(1,1) process:

$$y_t = \phi y_{t-1} + \varepsilon_t + \gamma \varepsilon_{t-1}$$

In lag operator notation, we have:

$$
\begin{aligned}
y_t &= \phi L y_t + \varepsilon_t + \gamma L \varepsilon_t \\
(1-\phi L) y_t &= (1+\gamma L)\varepsilon_t
\end{aligned}
$$

The multiplicative inverse of $(1-\phi L)$ is $(1+\phi L+\phi^2 L^2-\ldots)$, and so we can alternately express the ARMA process as an MA(∞):

$$
\begin{aligned}
(1+\phi L+\phi^2 L^2-\ldots)(1-\phi L) y_t &= (1+\phi L+\phi^2 L^2-\ldots)(1+\gamma L)\varepsilon_t \\
y_t &= (1+\phi L+\phi^2 L^2-\ldots)(1+\gamma L)\varepsilon_t \\
&= [(1+\phi L+\phi^2 L^2-\ldots)+(\gamma L+\gamma\phi L+\gamma\phi^2 L^2+\ldots)]\varepsilon_t \\
&= [\underbrace{1}_{\psi_0}+\underbrace{\phi^0(\phi+\gamma)}_{\psi_1} L-\underbrace{\phi^1(\phi+\gamma)}_{\psi_2} L^2-\underbrace{\phi^2(\phi+\gamma)}_{\psi_3} L^3+\ldots]\varepsilon_t \\
&= \psi(L)\varepsilon_t
\end{aligned}
$$

Assuming at every point that each step is well defined.

In this form, it is much easier to compute the various moments of the distribution.

Namely we have:

$$
\begin{aligned}
E[y_t] &= E\left[[1+\phi^0(\phi+\gamma)L+\phi^1(\phi+\gamma)L^2+\phi^2(\phi+\gamma)L^3-\ldots]\varepsilon_t\right] \\
&= [1+\phi^0(\phi+\gamma)L+\phi^1(\phi+\gamma)L^2+\phi^2(\phi+\gamma)L^3-\ldots]E[\varepsilon_t] \\
&= [1+\phi^0(\phi+\gamma)L+\phi^1(\phi+\gamma)L^2+\phi^2(\phi+\gamma)L^3-\ldots]0 \\
&= 0
\end{aligned}
$$

The variance of any $MA(\infty)$ process is given by:

$$
\begin{aligned}
\gamma_0 &= \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 \\
&= \sigma^2 \left(\psi_0^2 + \sum_{j=1}^{\infty} \psi_j^2\right) \\
&= \sigma^2 \left(1^2 + \sum_{j=1}^{\infty} (\phi^{j-1}(\phi+\gamma))^2\right) \\
&= \sigma^2 \left(1 + (\phi+\gamma))^2 \sum_{j=1}^{\infty} \phi^{j-1\,2}\right) \\
&= \sigma^2 \left(1 + (\phi+\gamma))^2 \sum_{j=0}^{\infty} \phi^{j\,2}\right) \\
&= \sigma^2 \left(1 + (\phi+\gamma))^2 \sum_{j=0}^{\infty} \phi^{2j}\right) \\
&= \boxed{\sigma^2 \left(1 + (\phi+\gamma))^2 \frac{1}{1-\phi^2}\right)}
\end{aligned}
$$

requiring that $\phi^2 < 1 \implies |\phi| < 1$

The jth autocovariance of the MA(∞) is given by:

$$
\begin{aligned}
\gamma_j &= \sigma^2 \sum_{k=0}^{\infty} \psi_k \psi_{j+k} \\
&= \sigma^2 \left(\psi_0 \psi_j + \sum_{k=1}^{\infty} \psi_k \psi_{j+k} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + \sum_{k=1}^{\infty} \phi^{k-1}(\phi+\gamma)\phi^{j+k-1}(\phi+\gamma) \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + \sum_{\kappa=0}^{\infty} \phi^{\kappa}(\phi+\gamma)\phi^{j+\kappa}(\phi+\gamma) \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \sum_{\kappa=0}^{\infty} \phi^{\kappa}\phi^{j+\kappa} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \sum_{\kappa=0}^{\infty} \phi^{2\kappa}\phi^{j} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \phi^{j} \sum_{\kappa=0}^{\infty} \phi^{2\kappa} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \phi^{j} \frac{1}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left(1 + (\phi+\gamma)\phi \frac{1}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left(1 + (\phi+\gamma) \frac{\phi}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left(\frac{1-\phi^2}{1-\phi^2} + \frac{\phi^2+\phi\gamma}{1-\phi^2} \right) \\
&= \boxed{\sigma^2 \phi^{j-1}(\phi+\gamma) \left(\frac{1+\phi\gamma}{1-\phi^2} \right)}
\end{aligned}
$$

Truly a bizarre expression that leaves little room for intuition. However, the series convergence there also only required that $\phi^2 < 1$, so $|\phi| < 1$ is still sufficient to ensure covariance stationarity.

2. Conditional Expectations

3. Conditional Likelihood

4. State Space Representation

A state space representation is a system of vector equations of the form:

$$
\begin{aligned}
y_t &= \mu + A z_t + H \eta_t \\
z_t &= \Phi z_{t-1} + R \xi_t
\end{aligned}
$$

Where $\eta_t \sim \mathcal{N}(0, I), \xi_t \sim \mathcal{N}(0, \Sigma)$. Letting $z_t = \begin{bmatrix} y_t & \varepsilon_t \end{bmatrix}'$ we can represent the observation equation as:

$$
y_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ \varepsilon_t \end{bmatrix} + 0\eta_t
$$

The η component will not be needed, so we can set $H = 0$. Here's a canonical way to represent the state equation that ensures dimensional conformability at every step.

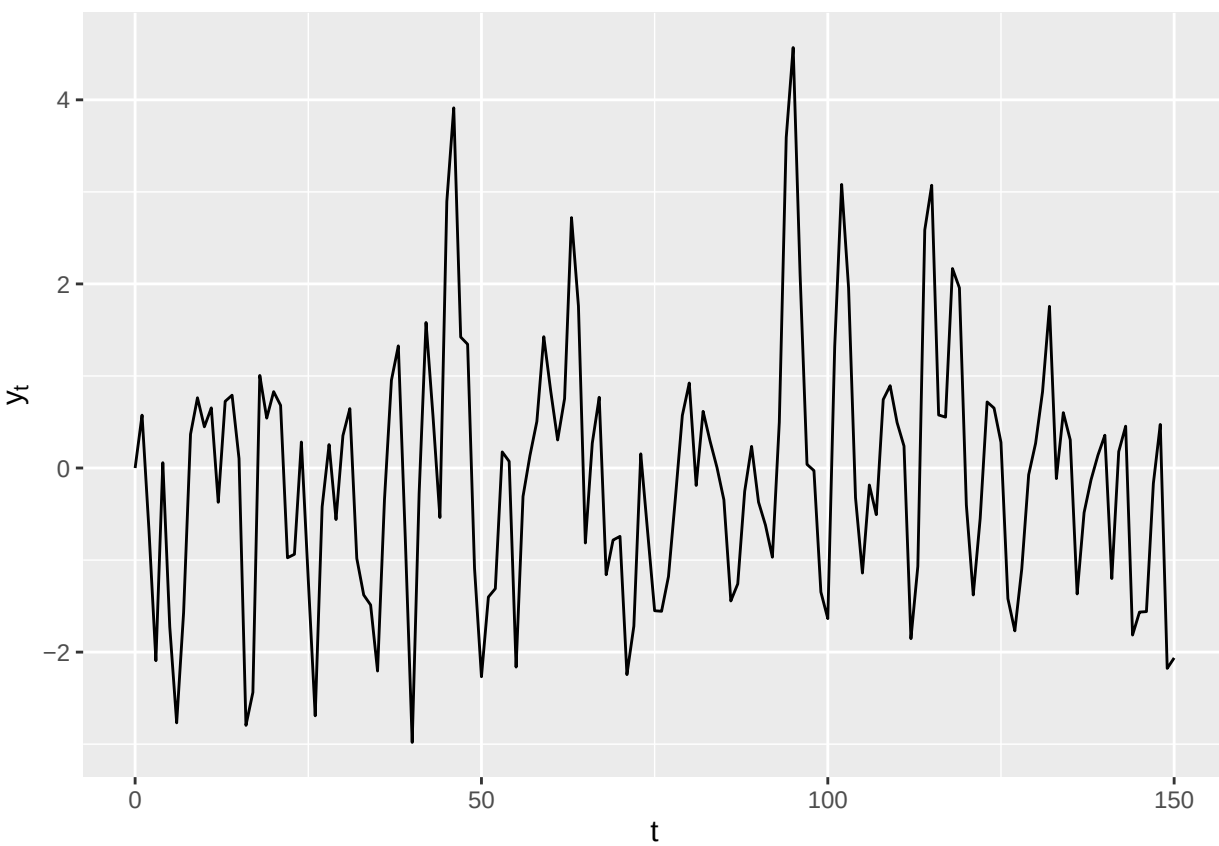
$$\begin{bmatrix} y_t \\ \varepsilon_t \end{bmatrix} = \begin{bmatrix} \phi & \gamma \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ \varepsilon_{t-1} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \xi_t$$

Where ξ_t is a 2-long vector distributed $\xi_t \sim \mathcal{N}(0, \sigma^2 I)$. You could also use a one dimensional ξ_t and then just the first column of the matrix before ξ_t.

5. Simulation

```
capT = 150
# structural parameters
phi = .9
gamma = .1
sigma_squared = 1

# state space parameters
dim_state <- 2
Sigma = sigma_squared * diag(nrow = dim_state)
R = matrix(c(1, 0,
             1, 0),
           byrow = T, nrow = dim_state)
Phi = matrix(c(gamma, phi,
               0, 0),
             byrow = T, nrow = dim_state)

# Observation parameters
dim_obs <- 1 #
mu <- c(0)
A <- matrix(c(1, 0), byrow = T, nrow = dim_obs, ncol = dim_state)
H <- matrix(c(0), byrow = T, nrow = dim_obs)

evolve_state <- function(z, Phi, R, Sigma){
  # compute multivariate normal using linearity
  xi = Sigma%*%rnorm(nrow(Sigma))

  (Phi %*% z) + R%*%xi
}

evolve_observations <- function(z, mu, A, H){
  eta <- rnorm(nrow(H))

  mu + (A %*% z) + H%*%eta
}

z0 = c(0,0)
simulate <- function(capT, z0, Phi, R, Sigma, mu, A, H){
  # loop capT times, feeding the result of the last iteration into
  # this this iteration
  states <- purrr::accumulate(
  1:capT,
  ~ evolve_state(.x, Phi, R, Sigma),
```

```r
  .init = z0
)

  # compute observation from latent state
  observations <- lapply(states, function(z){
    evolve_observations(z, mu, A, H)
  })

  # return long matrices with one row per time step
  states <- do.call(cbind, states) |> t()
  observations = do.call(cbind, observations) |> t()


  return(list(states = states, observations = observations))
}

sim_results <- simulate(capT, c(0,0), Phi, R, Sigma, mu, A, H)

library(ggplot2)
ggplot()+
  geom_line(aes(x=0:(nrow(sim_results$observations)-1),
                y=sim_results$observations[,1]))+
  labs(y = expression("y"[t]*""),
       x = "t")
```



6. Implementing the Kalman Filter

```
predict_y <- function(z, mu, A){
  mu + (A %*% z)
}

compute_v <- function(y, z, mu, A){
 y - predict_y(z, mu, A)
}

v_posterior_variance <- function(P, A, H){
  (A %*% P %*% t(A)) + H%*%t(H)
}

prediction_error_variance <- function(z, P, A){
  E
}
```

7. Applying the Kalman Filter

Metropolis Hastings

1. Numerical Maximization

2. Generating M-H Draws

3. Lots of Simulations